50 shades of informality: A gender analysis of informal labour and precarious working conditions

Abstract

This research challenges the widespread belief that the informal economy is an invisible and underreported phenomenon that cannot be analysed precisely since most of informal economic activities do not appear on official statistics. Using a unique dataset, this paper disaggregates Mexico's labour market into 50 economic activities that together employ around 98% of the national workforce. Hence, this is the first paper presenting estimates of informal employment rates across economic activities, including those assumed to be invisible parts of the informal economy. The analysis also shows how different forms of precarious working conditions are distributed across economic activities, demonstrating that labour precariousness is not exclusively experienced by workers with informal jobs. In addition, the paper finds that economic activities with high informal employment rates exhibit extreme gender segmentation—as women either predominate or are almost entirely absent—whereas activities with low informality rates display a more gender-balanced workforce and are closer to gender parity. Moreover, the paper finds that men are more likely to experience workload-related precarity, while women are more likely to experience income-related precarity, a pattern that may be reflecting internalised gender norms around breadwinning and housekeeping roles. Finally, the paper challenges the common assumption that women predominate in informal jobs. In sum, this paper represents a new advance in the study of the informal economy and can contribute to SDG 8 and the UN's Decent Work Agenda by showing that labour markets in developing countries require tailored policies to reduce labour precariousness according to the specific characteristics of each economic activity.

Keywords: Informal economy, gender division of labour, decent work

JEL Codes: J16, J21, J46, J81, O17

1 Introduction

Despite decades of study, this paper shows that we still lack a deeper understanding of the informal economy, of the analysis of precarious working conditions, and of how men and women experience these labour market challenges differently. The paper argues that incorrect assumptions, broad conclusions and ambiguous claims have shaped part of the literature on these topics. For instance, the informal economy is often described as difficult to analyse (Delechat and Medina, 2021; Ulyssea, 2020); there is no consensus on how to measure labour precarity (Canavire-Bacarreza et al., 2024; Burchell et al., 2014); and gendered analyses of these topics reach broad conclusions indicating that women are more likely than men to have informal, vulnerable or precarious jobs (Lo Bue et al., 2022; OECD, 2012; Chen, 2012).

Using a unique dataset, this paper introduces a novel analytical framework that disaggregates Mexico's labour market into 50 economic activities.[1] Based on this approach, the study offers new insights into these research topics by answering five interrelated research questions.

First, is the informal economy as complex to study as is often claimed? Several studies emphasise the challenges of measuring and studying the informal economy. Delechat and Medina (2021) argue that the informal economy "is a complex and multifaceted phenomenon that is difficult to measure and analyze". Ulyssea (2020) indicates that "the challenges of studying informality start with its measurement, since by definition informal activities are not officially registered". La Porta and Shleifer (2014) state that "measuring the informal economy is inherently difficult". Blades et al. (2011) also argue that "by its nature, the informal economy is difficult to measure". This is the first research challenging this claim, by showing how the informal economy can be effectively studied using labour statistics. It provides highly disaggregated estimates of informal employment rates and average labour income across 50 economic activities in Mexico (see online appendix, Figure 1.4). This granularity enables a deeper analysis of the informal economy, and it reaches an unprecedented level of detail that distinguish this research from previous studies.

Second, in which informal economic activities do men and women predominate? The literature on the gender division of labour argues that men hold a comparative advantage in agriculture and industry, while women have a comparative advantage in services (Galor and Weil, 1993; Ngai and Petrongolo, 2017). Nevertheless, these studies rarely consider that, in developing countries, formal and informal activities coexist across the three economic sectors. This research finds that informal activities tend to display extreme gender segmentation, with women either largely absent or constituting an overwhelming majority, depending on the activity. Meanwhile, formal economic activities are more likely to display a balanced gender workforce composition. Hence, the identification of formal and informal economic activities that are male-intensive, female-intensive, and

[1] *Mexico's labour market is disaggregated into 50 economic activities: 4 in agriculture, 17 in industry, and 29 in services. These economic activities are: (1) agriculture, (2) animal breeding and exploitation, (3) aquaculture, (4) fishing, (5) mining, (6) supply of electricity, water or gas, (7) informal construction, (8) formal construction, (9) formal manufacturing, (10) food industry, (11) beverage and tobacco industry, (12) textile inputs, (13) textile products (except clothing), (14) clothing manufacture, (15) leather goods manufacturing, (16) timber industry, (17) paper industry, (18) printing and related industries, (19) non-metallic mineral products, (20) metal products, (21) furniture, mattresses and blinds, (22) wholesale trade, (23) transportation by air, water, railways and roads, (24) media services, (25) finance and insurance, (26) real estate, (27) technical, professional and scientific services, (28) corporate management, (29) business support services, (30) education services, (31) health and social assistance, (32) culture, sport and leisure, (33) government, (34) street trade and street services, (35) Mexican grocery stores, (36) convenience and department stores, (37) retail: clothing, footwear and textiles, (38) retail: health-care products, (39) retail: stationery and other items, (40) retail: computers and household items, (41) retail: hardware and glassware, (42) retail: vehicles, auto parts and fuel, (43) internet- and catalogue-based retail, (44) temporary accommodation (hotels), (45) restaurants/bars, (46) repair and maintain cars and trucks, (47) repair and maintain machinery and appliances, (48) hairdressing and beauty services, (49) associations and organisations, and (50) domestic employees.*

closer to gender-parity represents a novel contribution to the literature on the gender division of labour (see Figure 1.1 in the section on descriptive statistics).

Third, are women more likely than men to have informal jobs? A large body of research claims that the informal economy is dominated by women.[2] Contrastingly, this research discusses new evidence that offers a more balanced picture. It highlights that men constitute the majority of the informal workforce in Mexico and worldwide, and that informal employment rates across countries are not higher for women than for men. The paper discusses various claims in the literature on this subject and advocates for more nuanced conclusions when studying women's participation in the informal economy.

Fourth, which precarious working conditions are more likely to be experienced by men and women in the labour market? Various studies have built composite indexes of labour precarity and concluded that women are more likely to have the most precarious and vulnerable jobs (Canavire-Bacarreza et al., 2024; Chen, 2012; Delechat and Medina, 2021; Lo Bue et al., 2022). By contrast, this paper takes a different approach by examining the likelihood that men and women experience different types of precarious working conditions. The results indicate that working women are more likely to experience income-related precarity: being unpaid workers, receiving daily wages, and having earnings below the minimum wage. On the other hand, working men are more prone to workload-related precarity, including working more than 48 hours per week, holding more than one job, working at night, and reporting feelings of underemployment.[3] Based on these results, the research suggests that these patterns may reflect that social expectations make men more likely to accept workload precarity, as this is intrinsically linked with assuming their breadwinner role, while women are more likely to accept income precarity because it is expected that they should prioritise bearing the historical burden of juggling domestic responsibilities, while also contributing to the household's financial situation.

Last, in which economic activities are workers more likely to experience different types of precarious working conditions? The analysis reveals that workers in residential construction are highly likely to work without a contract, despite the high physical risks of these jobs. Furthermore, workers in transportation services are highly likely to work more than 48 hours per week, while Mexican newspapers report that some drivers rely on cocaine to stay awake during excessively long shifts. Similarly, the analysis offers a granular view of how other types of precarious working conditions are distributed across Mexico's labour market.

For several reasons, Mexico offers a unique opportunity to study these topics within a unified framework. First, according to official statistics from 2025, Mexico is a developing country where more than 50% of the labour force has an informal job.[4] Second, this paper provides clear answers to the research questions thanks to a high-quality dataset. In 1994, Mexico signed the North American Free Trade Agreement (NAFTA) with Canada and the USA. These three countries adopted the North American Industry Classification System (NAICS) in their labour force surveys to compare business statistics. Hence, Mexico uses an advanced classification system that can identify the

[2] *My research has identified at least 15 studies published between 1996 and 2024 suggesting that women predominate in the informal economy. Some of these studies have been highly influential on the study of gender and informality. Their arguments converge on three main claims: (1) women are over-represented in the informal sector worldwide; (2) women exhibit higher informal employment rates than men across countries; and (3) women are more likely than men to have informal jobs. The literature review section presents more details on these studies.*

[3] *Underemployment refers to individuals who are working but report that they would like to be working more hours.*

[4] *In Mexico, labour informality is measured based on social security registration. Wage employees are classified as informal if their employer does not register them with the Mexican Institute of Social Security (IMSS). The self-employed are considered formal only if they pay their own pension contributions, have access to medical services and they have formal accounting records. (See Table 1.8 and 1.9 in the online Appendix for more information about the informal workforce in Mexico).*

activities in which workers are employed, even when their jobs are informal. Third, the ENOE survey includes a set of questions that are well-suited to analysing different types of precarious working conditions that extend beyond the formal-informal dichotomy. Hence, Mexico offers a unique setting to study the informal economy and different forms of precariousness at a level of disaggregation not previously available in other studies.

The research suggests that other countries may replicate the methodology and empirical strategy presented in this analysis to study informal labour markets and precarious working conditions. The key requirements are that their datasets can disaggregate the labour market into different economic activities, or different occupational groups, and that the estimates remain within acceptable coefficients of variation. This entails consideration of the sampling design, the primary sampling unit, the sampling weight, and the overall structure of the labour force survey.

This paper also has important policy implications for the UN Decent Work Agenda, as the findings illustrate that efforts to reduce informality and precariousness in the labour market cannot rely on one-size-fits-all approaches. Instead, effective interventions must rely on tailored policies that consider the heterogeneity of economic activities documented in this research. The remainder of the paper is organised as follows: Section 2 situates the study within the literature, Section 3 details the methodology, Section 4 presents the results, Section 5 presents a brief discussion of women in the informal economy, and Section 6 offers the concluding remarks.

2 Literature Review

This section is divided into five subsections. It begins by presenting foundational theories on the informal economy and examining how informality is currently measured. It then discusses theories and evidence on gender patterns in the division of labour. Next, it explores the relationship between gender, informality and labour precarity. It also highlights the lack of consensus on how to measure labour precarity. The following subsection discusses how the analytical framework of this research has been inspired by various strands of literature that use labour market disaggregation as an empirical strategy to generate new insights about work patterns.

2.1 Informality

Two foundational economic theories laid the groundwork for understanding the existence of informal economic activities in the labour market. Lewis (1954) explained that in developing countries two sectors coexist: a large subsistence sector, where workers are engaged in agriculture, petty services, and other casual informal jobs with near-zero marginal productivity, and a small but emerging modern industrial sector that is capital-intensive and has higher productivity rates. Since the latter expands only through reinvested profits, its capacity to absorb labour is limited. As profits are reinvested, however, this absorption capacity gradually increases, drawing workers out of informal subsistence jobs into more productive employment in the modern industrial sector. Later on, Harris and Todaro (1970) proposed a dual-economy model centred on the coexistence of rural and urban labour markets. They modelled workers' migration decisions from rural labour markets with low wages to urban labour markets offering higher remuneration. The model anticipated that individuals migrate based on expected wages rather than actual wages, and that the inflow of migrants to urban areas typically exceeds the number of formal jobs available. As a result, an urban informal sector emerges as a safety net, absorbing migrants into precarious, unregulated employment while they await opportunities in the formal economy.

The literature has also identified four theoretical schools that propose different explanations for the existence of informal labour markets (Chen, 2012). The dualist school argues that the informal economy emerges from a mismatch between population growth and modern industrial employment, alongside asymmetries between workers' skills and job requirements (International Labour Organization, 1972; Hart, 1973). The structuralist school conceptualises the informal economy as a capitalist strategy of large firms that rely on subordinated economic units and precarious forms of labour to reduce labour costs and enhance competitiveness (Castells and Portes, 1989; Moser, 1978). The legalist school argues that companies and workers operate outside regulatory frameworks to avoid the high costs and complexities of formal registration (de Soto, 1989). Finally, the voluntarist school argues that being informal is a rational decision where individuals make a cost benefit analysis to decide whether to participate in the formal or informal sector (Maloney, 2004).

Building on these conceptual foundations, attention turned to the challenge of defining, identifying and measuring the informal workforce. Early approaches relied on the 'enterprise definition', which classified workers according to the characteristics of the firm they worked for. Under this method, informal workers were those who were self-employed or employed in small firms with fewer than five workers (Chen, 2012; International Labour Organization, 2018; Khamis, 2012; Maloney, 1999). This threshold was established because firms with few workers rarely register with tax or social security authorities, while often operating in cash and without standard accounting records. Hence, the enterprise-based definition was used to measure the size of the 'informal sector', a statistical category adopted in 1993 by the International Conference of Labour Statisticians (ICLS). However, the informal sector definition has clear limitations, as it excludes informal labour within firms employing more than five workers and formal enterprises hiring workers 'off the books' (Chen, 2012; Ulyssea, 2020). For instance, Bosch (2006) reports that about 30% of jobs in medium-sized firms (11–50 workers) and 8% in large firms (over 250 workers) are informal.

The failure of the enterprise-based measurement of informality prompted a shift towards a broader job-based conceptualisation. This led to a methodology measuring 'informal employment' by job characteristics rather than by firm characteristics. This expanded statistical definition focused on identifying jobs that lack access to social protection, both inside and outside informal enterprises (Chen, 2012). Based on this discussion, the ILO (International Labour Organization, 2013) argues that the terms 'informal sector' and 'informal employment' are not synonyms and should not be used interchangeably, since the specific distinctions between each term are critically important from a technical point of view.

In the case of Mexico, Khamis (2012), Conover et al. (2022), and Samaniego de la Parra and Fernández Bujanda (2024) explain how informality is measured in the country. Wage employees are classified as informal if their employer does not register them with the Mexican Institute of Social Security (IMSS), as required by law. This registration gives them access to employment benefits, including a pension scheme, healthcare, daycare, disability insurance, and maternity and sick leave. Accordingly, IMSS deducts payroll taxes from employers and employees to fund these benefits. In contrast, Ulyssea (2018) explains that informality in Brazil is measured simply by checking whether employees have a written contract or not. This example thus illustrates that the measurement of labour informality in Mexico is more meticulous.

INEGI (2014) offers a detailed methodological note on how it identifies informal workers in Mexico. The institute explains that the methodology draws on an integrated conceptual framework developed in collaboration with the International Labour Organization (ILO) through the Delhi

Group (a group of experts on informal sector statistics). This approach combines two core perspectives based on the Hussmanns Matrix. For independent workers (employers and self-employed), Mexico applies the enterprise-based framework for the informal sector. For instance, if independent workers perform economic activities without keeping any formal accounting records, they are considered part of the informal workforce. As evidence, the ENOE survey (Encuesta Nacional de Ocupación y Empleo) fourth quarter of 2019 reports that only 14% of the self-employed are considered formal workers in Mexico.

On the other hand, INEGI (2014) explains that subordinate workers are classified as informal if they report that their employer is not providing them with access to medical services at the Mexican Institute of Social Security (IMSS). This is the question through which INEGI identifies whether the employer of that subordinate employee is paying payroll taxes to provide them with access to social protection, given that IMSS is responsible not only for providing access to health services, but also for access to daycare, pensions, sick leave and maternity leave, among other formal job benefits. Hence, this dual approach enables Mexico to capture informality across different types of labour arrangements.

2.2 The gender division of labour

Different economic theories have tried to explain the gender division of labour by discussing men's and women's comparative advantages both in the labour market and within the household. Becker (1985) presents an energy allocation model to show how the effort-intensive nature of housework, particularly childcare, reduces women's residual energy to engage in paid work. This reduction lowers productivity, depresses hourly wages and incentivises shifts towards less physically demanding occupations. Furthermore, Becker (1993) argues that biological differences (such as women's capacity to bear, give birth to and breastfeed children) generate distinct comparative advantages in the labour market. As a result, women specialise in domestic production, while men specialise in market-oriented work. The author argues that this pattern reflects a household maximisation strategy under a single joint utility function and helps to explain the historical pattern of men assuming breadwinner roles while women concentrate on domestic production. Meanwhile, Galor and Weil (1993) propose another model to explain the gender division of labour. In this case, men are endowed with more raw physical strength, while both sexes have equal endowments of mental capacities. This asymmetry creates a comparative advantage for men in physically intensive occupations, leading to their predominance in such roles. Nevertheless, as an economy develops and accumulates physical capital, the relative return to mental labour rises, shifting demand away from physical tasks and thus increasing female labour participation rates.

In line with this theory, empirical evidence indicates that men have a comparative advantage in goods-producing sectors that rely more heavily on physical strength (e.g. agriculture, construction, manufacturing), while women have an advantage in services that demand intellectual skills (Ngai and Petrongolo, 2017). Evidence also shows that occupations requiring interpersonal skills employ more women than men, and the rise of people-oriented tasks has boosted female labour participation (Borghans et al., 2014). Based on psychological and neurological research, the authors argue that women perform better in these tasks because they place greater weight on the impact of their actions on others, as women exhibit higher levels of empathy than men. There is also evidence indicating that women have benefited from the expansion of low-skill services such as wholesale trade, restaurants, hotels and retail stores (Nayyar et al., 2021). Finally, an analysis of gendered patterns in the US labour market between 1950 and 2019 categorised jobs into three types of skills:

cognitive (brain), physical strength (brawn) and coordination (motor). The findings reveal a decline over time in demand for brawn-intensive labour alongside a rise in demand for cognitive skills, and conclude that this shift from 'brawn' to 'brain' tasks has strengthened women's comparative advantage in the labour market (Rendall, 2024).

Conventional theories about the gender division of labour explain why men have historically specialised in market work and women in home production. The gendered analysis of comparative advantages in the labour market also clarifies why women participate more in services than in manufacturing or agriculture. Yet these theories fail to explain a persistent feature of modern households: despite the expansion of services in developed countries and women's rising hours in paid employment, they continue to bear a disproportionate share of housework. Bertrand et al. (2015) find that in couples where the wife earns more than the husband, the wife still spends more time on domestic chores. The previous theories would predict that, in this scenario, household tasks would be shared equally between husband and wife; however, this is not the case.

Akerlof and Kranton (2000) developed a model of household behaviour to explain the disparities in the division of household labour by incorporating 'identity' and 'self-image' into the household utility function. They argue that societal prescriptions, internalised by men and women, create identity-based payoffs and result in utility losses if they are not fulfilled. These prescriptions include the belief that husbands should earn more than their wives and that men should not do domestic work. Hence, wives may undertake a larger share of housework even if they work similar hours to their husbands, or even if they are the primary income earners. The authors use this model to illustrate how gender norms can influence the gender division of labour within the household.

Kanji and Samuel (2017) offer evidence of how gender roles may shape men's labour market behaviour. They find that, in Western Europe, men who embrace traditional breadwinner roles and endorse conservative views on women's employment are more likely to report feelings of underemployment[5]. Surprisingly, these conservative men want to work more hours, even when already working substantially longer than men with egalitarian attitudes toward female labour participation. This paradox suggests that feelings of underemployment may stem from internalised expectations about fulfilling the breadwinner role.

For Mexico, Cunningham (2001) argues that household roles, rather than gender, exert a stronger influence on patterns of female labour supply. Single women without children display labour supply behaviours closer to those of men than to those of married women, as they are more likely than any other group to hold inflexible, higher-paying positions in the formal sector. Conversely, women with children tend to take flexible, informal jobs to reconcile domestic responsibilities with labour market participation.

Finally, Gottlieb et al. (2024) examine cross-country differences in how men and women allocate time between market work, unpaid domestic tasks, and unpaid care work. A key finding is that men and women may derive differing levels of utility or disutility from engaging in the same activities. Specifically, market work generates higher disutility for women, whereas men exhibit greater disutility from non-market activities[6]. They argue that women may experience greater disutility from market work as a result of exposure to sexual harassment, chemical hazards during pregnancy or safety concerns linked to late-night shifts.

[5] *Underemployment refers to workers who report wishing to work more hours than they currently do.*

[6] *They define market work as any activity that produces goods or services falling within the System of National Accounts (SNA), including subsistence agriculture, fetching water, or collecting firewood. Non-market activities have two specific subcategories: domestic chores and unpaid caregiving.*

Taken together, these theories — from comparative advantage and identity economics to gender norms and household roles — provide the analytical foundation for interpreting the gendered labour market patterns documented in the rest of this analysis.

2.3 Gender and informality

Several studies have identified factors that make women more likely to take informal jobs. Motherhood increases the likelihood of informality among women, while fatherhood does not affect men's probabilities (Berniell et al., 2021; Ulyssea et al., 2025). Gender roles assign women primary responsibility for household and care work, creating a need for informal, flexible jobs to balance domestic chores with paid labour (Delechat and Medina, 2021; Maloney, 2004). The current arrangements of most labour markets are based on the old family role model, where men access formal jobs with higher earnings and better benefits because they are considered the primary household providers (Sabates-Wheeler and Kabeer, 2003). Collectively, these factors have been identified as contributing to women's participation in informal employment.

Similarly, this research identified at least 15 studies published between 1996 and 2024 that portray the informal economy as a space dominated by women. Many of these studies have been highly influential in the field of gender and informality. Their arguments converge on three main claims: 1) women are over-represented in the informal sector worldwide; 2) women exhibit higher informal employment rates than men across countries; and 3) women are more likely than men to have informal jobs. The following passage provides direct citations to these studies.

Funkhouser (1996) claims that "females are significantly more likely to be employed in the informal sector". Sethuraman (1998) states that "women in all age groups depend on the informal sector more heavily than men". Chen (2001) claims that "women are over-represented in the informal sector worldwide". Losby et al. (2002) state that "women are more likely than men to work informally". Gallaway and Bernasek (2002) note that, in developing countries, the workforce is mostly in the informal sector but "women in particular are disproportionately represented in this sector". Maloney (2004) argues that, in Mexico and other Latin American countries, there is a "disproportionate representation of women in informal self-employment". Chant and Pedwell (2008) discuss the "predominance of women in the informal economy". The OECD (OECD, 2012) states "that women are more likely than men to work informally". McCaig and Pavcnik (2015) say that "women are more likely to work in the informal sector". The OECD (OECD, 2019) mentions that "women are over-represented in the informal workforce in a majority of countries". Malta et al. (2019) state "that women are more likely to be in the informal sector than men". Delechat and Medina (2021) argue that "in two out of three low- and lower-middle income countries, women are more likely than men to be employed informally". Berniell et al. (2021) state that in Latin America, "informality rates are higher for women than for men". Gardner et al. (2022) argue that "in more countries the share of women in informal employment exceeds that of men". The OECD (OECD, 2023) notes that "informal employment is a greater source of employment for women than for men in the following regions: sub-Saharan Africa, Central and Western Asia, Southern Asia and in Northern, Southern and Western Europe". Finally, Ortiz-Ospina et al. (2024) state that, in most countries, "women tend to work in the informal economy more often than men".

All the aforementioned studies converge on a narrative that frames the informal economy as a space dominated by women. Nevertheless, some studies have presented contrasting evidence. Angel-Urdinola and Tanabe (2012) argue that in the Middle East and North Africa (MENA), women are less likely than men to work informally, mainly because these countries have unusually large public sectors. Assaad and Barsoum (2019) have documented that public sector jobs in these countries account for some of the highest shares worldwide: Jordan (40%), Saudi Arabia (38%), Algeria (31%) and Egypt (25%). They also note the high share of public sector jobs in China (29%) compared with the smaller shares in Indonesia (8%), Turkey (15%) and Mexico (15%). The central argument of these studies regarding the MENA region is that women in these countries prefer not to work rather than to work informally. Consequently, the few women who participate in the labour market are formally employed in public sector jobs. The ILO (International Labour Organization, 2023) also notes that in China and Russia "men face greater exposure to informality", although without attributing this pattern to the size of their public sector. In addition, an earlier report from the ILO (International Labour Organization, 2018) presents statistics that cast doubt on several of the claims regarding women's predominance in informality. This report indicates that, globally, informal employment rates are higher for men (63%) than for women (58%). It also states that there are two billion informal workers worldwide, of whom roughly 740 million are women and 1,240 million are men, which means that globally, 37% of informal workers are women, while 63% are men. Hence, although there once appeared to be a consensus on women's predominance in the informal economy, new statistics are challenging this claim.

2.4 Gender and labour precarity

Various studies have also concluded that women are more likely than men to have precarious and vulnerable jobs, and they have listed different factors to explain this pattern (Canavire-Bacarreza et al., 2024; Chen, 2012; Delechat and Medina, 2021; Lo Bue et al., 2022). It is argued that marriage raises women's likelihood of holding vulnerable jobs, as gendered social norms often assign them primary household responsibilities, making precarious work more compatible with these constraints. The evidence also indicates that motherhood further channels women into vulnerable roles, as a result of heavier childcare demands. Additionally, women's greater likelihood of being unpaid workers inevitably increases their exposure to precarious jobs. However, gender studies on precarious working conditions conclude with broad findings that fail to examine men's outcomes in depth, underscoring the need for a more granular examination of this research topic.

In another strand of the literature, studies on decent work, labour precarity, job quality and vulnerable employment have focused extensively on methodological debates about how to measure these concepts under a standardised framework. These studies share a common concern: assessing the quality, security and stability of employment. However, differences in concepts, methodologies and indicators often fail to generate robust or consistent measures. Burchell et al. (2014) criticise the parallel development of job quality measures and Decent Work indicators. They argue that Decent Work is a vague and broadly defined concept, almost impossible to measure across countries. In contrast, they consider that the measurement of job quality has progressed more consistently in generating comparable cross-country data.

Canavire-Bacarreza et al. (2024) provide an insightful overview of job-quality measurement. They highlight the absence of universal standards and instead note the coexistence of diverse methodologies used by international organisations and researchers. For instance, between 2008 and 2009, the ILO introduced statistical indicators covering several dimensions of the Decent Work Agenda, including adequate earnings, working time, job stability and other key factors. Meanwhile, the OECD developed a Job Quality Framework, which assesses earnings, employment security and working environment. Thus, each organisation uses its own methodology rather than working towards a common standard.

Canavire-Bacarreza et al. (2024) also discuss scholars' methodologies to measure job quality. While some apply Principal Component Analysis (PCA), others use the Alkire-Foster multidimensional approach. Furthermore, because of differences in data availability across countries, the measurement of job quality in these studies is based on different variables[7]. Overall, these studies show the absence of a unified framework and a lack of consensus over how job quality and labour precarity should be defined and measured, as they use different variables, techniques and indicators according to its own criteria and data constraints (Brummund et al., 2018; Canavire-Bacarreza et al., 2024; Conover et al., 2022; Del Carpio et al., 2017; Hovhannisyan et al., 2022; Lo Bue et al., 2022; Sehnbruch et al., 2020; Yañez, 2018).

Finally, in the case of Mexico, the Ministry of Labour presents a series of indicators to study different types of precarious working conditions in the country (Secretaria del Trabajo y Prevision Social, 2022). These indicators include being an informal worker, excessive working hours, underemployment, having two jobs, not having a contract, having a temporary contract, income below the minimum wage, access to healthcare, access to a pension scheme, access to paid annual leave, access to an end-of-year bonus (also known as *aguinaldo*), access to profit-sharing at the end of the fiscal year, access to savings for housing and child labour, among others.

Therefore, this review shows that studies on labour precarity employ different terms, methodologies and variables. Despite this, most studies conclude that women hold the most precarious jobs in the labour market. To make a novel contribution to this research topic, this paper examines different types of precarious working conditions separately to avoid methodological debates about weighting and indicators that should be considered to build a composite index. Instead, it analyses labour precarity as the multidimensional phenomenon it truly is.

2.5 Analytical frameworks based on labour market disaggregation

This subsection begins by discussing the analytical frameworks used to study informal employment and labour precarity, many of which rely on labour market disaggregation to capture different employment patterns. Perry et al. (2007) propose an analytical framework to distinguish between two fundamental drivers of informality: exit and exclusion. Voluntary 'exit' is used to identify workers or firms that choose to operate informally. In contrast, involuntary 'exclusion' is used to identify workers or firms pushed into the informal economy thanks to systemic barriers. Subsequent studies have proposed other frameworks to study the informal economy.

Ulyssea (2018) present another analytical framework where firms can exploit two margins of informality. First, the extensive margin (whether they register their business or not). Second, the intensive margin (formal firms choosing between hiring workers 'off the books' or not). This framework is useful for illustrating that formal enterprises can still hire informal workers and shows that, in Brazil, at least 40% of the informal workforce is employed by formal firms. Furthermore, Conover et al. (2022) adopt an analytical framework that disaggregates Mexico's labour market

[7] *The cited studies have employed a wide range of variables to measure job quality: 1) adherence to labour law regulations, working conditions, productive use of skills, employment resilience, career opportunities, linkages between wages and jobs; 2) indicators of access to labour rights such as health insurance, retirement benefits, unemployment insurance, pension schemes, annual paid leave, and paid sick leave; 3) indicators of employment stability including tenure, contract type (written, permanent, temporary), labour stability, and occupational status; 4) indicators of job attributes such as job satisfaction, bonuses, paid vacation days, and career opportunities; 5) income-related variables including labour income, sufficient income to live out of poverty, minimum wage compliance, relative underpayment, and overqualification; and 6) working-time variables such as weekly hours worked, excessive working hours, unemployment, underemployment (desire to work more hours), number of jobs, and having a second job.*

into three job categories: (1) formal salaried, (2) informal salaried and (3) self-employed. Their analysis concludes that informal salaried jobs are consistently of lower quality than both formal salaried and self-employed jobs.

Chen et al. (2006) criticise studies comparing informal self-employment with informal wage employment. Instead, they propose using more detailed disaggregation frameworks to account for the diverse forms of informal jobs. Subsequently, Chen (2012) propose an analytical framework that disaggregates the informal labour market into six categories: (1) informal employers, (2) regular informal wage workers, (3) informal self-employed, (4) casual informal wage workers, (5) industrial homeworkers and (6) unpaid family workers[8]. This framework leads to the conclusion that women are typically concentrated in the most precarious forms of informal employment (such as unpaid family workers), while men are considerably more likely to be informal employers.

A key shortcoming of these studies of the informal economy is their limited level of disaggregation, as their analytical frameworks fail to capture the heterogeneity of informal labour markets, suggesting the need for an alternative approach with finer disaggregation to generate deeper insights. In this regard, the literature on structural transformation offers a useful framework for disaggregating labour markets. Initially, such studies divided the labour market into three broad categories: agriculture, industry and services. However, more recent research has shifted to a higher degree of disaggregation, yielding deeper insights about industry and services. McMillan et al. (2014) analyse patterns of structural transformation by disaggregating the labour market into nine sub-sectors[9]. Furthermore, Timmer and de Vries (2009) as well as van Neuss (2019) use a 10-sector dataset to study structural transformation in Asia, Latin America, Africa and Europe. Finally, Ngai and Petrongolo (2017) study structural transformation and the rise of the service economy by disaggregating the labour market into 17 sub-sectors[10].

Other studies analyse labour markets by focusing on occupations and tasks, rather than on sub-sectors of the economy. Rendall (2024) study the US labour market by analysing 300 occupations and 12,000 tasks, providing a framework to analyse the relative importance of brain, brawn and motor skills across different jobs. Autor and Dorn (2013) document the recent growth of low-skill service jobs in the US labour market by analysing the task content of more than 300 occupations, while Acemoglu (2025) analyse more than 4,000 tasks to estimate the macroeconomic implications of Artificial Intelligence in the US labour market. Together, these studies reflect a growing shift in labour market research towards classifications that go beyond traditional sectoral boundaries, enabling a deeper understanding of employment structures.

Hence, this paper argues that informality is highly heterogeneous. Classifying workers as either informally self-employed or informally wage employed does not capture this complexity. Likewise, disaggregating the labour market into ten broad economic sub-sectors lacks the granularity required

[8] *Chen (2012) divides the informal workforce into six categories. Informal employers are entrepreneurs running informal businesses that hire workers. Informal self-employed are running informal businesses without hiring workers. Regular informal wage workers are employees hired by enterprises without social protection benefits such as pensions or medical coverage. Casual informal workers face a daily risk of losing their jobs or earnings. Industrial homeworkers produce goods from home as piece-rate workers. Unpaid family workers perform activities in family businesses without receiving any payment.*

[9] *These nine sub-sectors are: 1) agriculture, hunting, forestry and fishing; 2) mining and quarrying; 3) manufacturing; 4) public utilities (electricity, gas and water); 5) construction; 6) wholesale trade, retail trade, hotels and restaurants; 7) transport, storage and communications; 8) finance, insurance, real estate and business services; and 9) community, social, personal and government services.*

[10] *These 17 sub-sectors are: 1) agriculture, forestry, fishing, mining and extraction; 2) construction; 3) manufacturing; 4) utilities; 5) transportation; 6) post and telecommunications; 7) wholesale trade; 8) retail trade; 9) finance, insurance and real estate (FIRE); 10) business and repair services; 11) personal services; 12) entertainment; 13) health; 14) education; 15) professional services; 16) welfare and nonprofit activities; and 17) public administration.*

for meaningful analysis. Furthermore, disaggregation by hundreds of occupations or thousands of tasks hinders the accurate estimation of informal employment rates. Therefore, this study adopts an intermediate level of disaggregation by categorising Mexico's labour market into 50 economic activities, enabling a granular and reliable estimation of informal employment rates.

3 Methodology

This section is divided into three parts. The first presents an overview of the dataset. The second presents the econometric model's details. The third presents the descriptive statistics and also includes all the details of the methodology for disaggregating the Mexican labour market into 50 economic activities. Taken together, these sections outline the empirical strategy of this research.

3.1 Dataset details

The ENOE survey is the country's largest statistical project and the main source of labour market information. It is representative at the national level and follows a rotating panel design. Hence, individuals are interviewed for five consecutive quarters, and 20% of the sample is replaced each quarter. To avoid repeated observations and to capture seasonal variation across different years, this study uses four non-overlapping rounds: Q1 2016, Q2 2017, Q3 2018, and Q4 2019.

The ENOE survey conducted by the National Institute of Statistics and Geography (INEGI) includes two variables that classify people in the labour force based on the NAICS. This classification system was adopted by Mexico, the USA and Canada as part of NAFTA, signed in 1994. The system disaggregates Mexico's economic sectors into 20 sub-sectors[11]: one in agriculture, four in industry, and fifteen in services. In addition, the NAICS classification system in the ENOE survey contains a secondary variable with 181 distinct codes for economic activities within these sub-sectors. Consequently, each of the 20 sub-sectors possesses a more granular classification of the constituent economic activities. Using these 181 codes, the Mexican labour market was disaggregated into 50 economic activities. The criteria and methodology for this disaggregation, along with descriptive statistics, are presented in the online Appendix.

Additionally, the ENOE contains a series of variables that the Mexican Ministry of Labour uses to analyse different precarious working conditions in the country (Secretaria del Trabajo y Prevision Social, 2022). Based on these data, the research explores nine types of precarious working conditions in Mexico's labour market: (1) being an informal worker; (2) being a subordinate worker without a written contract; (3) working more than 48 hours per week; (4) being an unpaid worker; (5) having a daily wage; (6) having more than one job; (7) working during non-daytime hours; (8) reporting feelings of underemployment; and (9) earning less than the minimum wage.

The Mexican Ministry of Labour also monitors whether Mexican employees have access to other labour benefits (medical services, pension, paid annual leave with full salary, year-end profit-sharing arrangements, Christmas bonus, and housing credit schemes). Workers registered with the Mexican Social Security Institute are legally entitled to these benefits under formal employment

[11] *The 20 sub-sectors are: 1) agriculture, 2) mining, 3) generation and distribution of electricity, gas and water, 4) construction, 5) manufacturing, 6) wholesale trade, 7) retail trade, 8) transportation services, 9) mass media, 10) financial services, 11) real estate, 12) professional, scientific and technical services, 13) corporate services, 14) business support, 15) education, 16) health, 17) entertainment and cultural services, 18) accommodation and food services, 19) government and 20) other services.*

provisions in Mexican labour law. Hence, rather than focusing on labour benefits intrinsically linked to formal employment status, this research examines variables that capture distinct and relatively independent aspects of precarious working conditions in the labour market.

3.2 Econometric model

The econometric analysis presented in this study has three objectives. First, to evaluate the hypothesis that women are more likely to be informal workers than men. Second, to examine which precarious working conditions are more likely to be experienced by men and women in the labour market. Third, to identify which precarious working conditions are more likely to be observed across the 50 economic activities considered in this analysis.

To achieve these objectives, this research executes nine probit regressions using nine binary dependent variables: (1) being an informal worker; (2) being a subordinate worker without a contract; (3) working more than 48 hours per week; (4) being an unpaid worker; (5) receiving a daily wage payment; (6) having more than one job; (7) working during non-daytime hours; (8) reporting feelings of underemployment; and (9) earning less than the minimum wage. The details of these binary variables are explained in the following subsections.

It should be noted that the research employs separate probit regressions rather than a multivariate probit regression for computational reasons. Cappellari and Jenkins (2003) developed the command to execute multivariate probit regressions and highlighted a significant computational limitation: datasets with several thousand observations require weeks of processing time when the analysis includes more than six dependent variables. Given the computational burden of estimating a multivariate probit with nine equations and over 600,000 observations, this research employs separate probit regressions instead.

All the dependent variables, including labour informality, are considered as different types of precarious working conditions by the Mexican Ministry of Labour, which uses them to assess precarious working conditions in the country (Secretaria del Trabajo y Prevision Social, 2022). Thus, the nine binary dependent variables employed in this analysis are defined as follows:

1. Being an informal worker: Equal to 1 if the individual is part of the informal workforce, 0 if they are part of the formal labour force.

2. Being a subordinate worker without a contract: Equal to 1 for employees without a labour contract, 0 for employees with a contract.

3. Being an unpaid worker: Equal to 1 if the individual is working without getting paid, 0 if they work and receive any remuneration.

4. Being paid on a daily basis: Equal to 1 if the respondent reports receiving daily payments, 0 if they are paid in any other payment schedule.

5. Earning less than the minimum wage: Equal to 1 if the individual works more than 48 hours per week but earns less than the minimum wage; equal to 0 for all other individuals in the labour force who do not meet both conditions.

6. Working more than 48 hours per week: Equal to 1 if the respondent reports working over 48 hours per week, 0 if they report working less than 48 hours.

7. Having more than one job: Equal to 1 if the individual has more than one job, 0 if they only have one job.

8. Working during non-daytime hours: Equal to 1 if the respondent reports that part or all of their working schedule is not between 6 am and 8 pm; equal to 0 if the respondent's working schedule is between 6 am and 8 pm.

9. Reporting feelings of underemployment: Equal to 1 if the worker reports a willingness or need to work more hours than they are currently working; equal to 0 if the individual is part of the workforce but does not report such willingness or need to work more hours.

The econometric specification is identical across the nine regressions, with the only difference being the dependent variable employed in each case. Hence, the explanatory structure of the model maintains the same specification across regressions. These regressions will show the probability of men and women experiencing nine different precarious working conditions[12]. They do so while controlling for individual and household characteristics, as well as for the economic activity in which each individual in the sample is employed. Hence, the specification of the econometric model is expressed in the following equation:

$$\begin{aligned} P(Y_{i,p} = 1) = \Phi\Big(&\beta_0 + \beta_1 Female_i + \beta_f EconomicActivity_i \\ &+ \sum_{j=4}^{j=1} \beta_j Ind_i^{(j)} + \sum_{k=4}^{k=1} \beta_k Hou_i^{(k)} + \mu_{s,t}\Big) \end{aligned} \tag{1}$$

'Y' represents the nine dependent variables that capture different types of precarious working conditions. Therefore, $P(Y_{i,p} = 1)$ denotes the probability that an individual (i) is experiencing a specific dimension of labour precarity (p). The dependent variables take the value of one if the individual (i) is experiencing that precarious working condition (p), and 0 if they are not experiencing it. Φ represents the standard normal cumulative distribution function of the probit regressions, and β_0 is the intercept.

A core explanatory variable in these regressions is 'Female', which takes the value of 1 for women and 0 for men. Hence, β_1 is the main parameter of interest. A positive and statistically significant coefficient on this variable implies that women are more likely than men to experience a specific precarious working condition. Conversely, a negative and statistically significant coefficient would indicate that men are more exposed to that dimension of labour precarity.

The other core explanatory variable of the model is 'Economic Activity'. In this case, $f \in \{1, \ldots, 50\}$ is an index to represent each of the fifty economic activities. Therefore, the coefficients β_f will indicate whether workers employed in each of these fifty activities are more likely to experience a specific type of precarious working conditions. A positive and statistically significant coefficient suggests that this dimension of labour precarity is more prevalent among workers employed in that activity. The next subsection details how this variable was constructed.

The analysis also includes a set of control variables. 'Ind' represents a vector of four control variables (j) capturing individual characteristics. Age and age squared are included to model the potential U-shaped relationship between age and informality, since prime-age workers are typically

[12] *The research considers that being an informal worker (in contrast to a formal worker) is only one dimension of precarious working conditions. The other eight variables illustrate that labour precarity extends beyond informality and recognises the existence of multiple types of precarious working conditions in the labour market.*

more likely to have formal jobs. Marital status is also included as a control, as evidence suggests that marriage makes women more likely to take informal and precarious jobs. Finally, education is incorporated to account for the strong association between educational attainment and job quality. Furthermore, 'Hou' represents a vector of four control variables (k) capturing household characteristics. First, a categorical variable indicating whether households belong to one of four socioeconomic strata: high, medium-high, medium-low or low. INEGI constructs this variable based on the presence or absence of 34 types of household assets. Second, a variable measuring the number of household members to control for household size. Third, a binary variable indicating whether the household is in a rural or urban area. Finally, as an additional control, a continuous variable showing the number of children under the age of six residing in the household.

The fixed effects included in the model are denoted by μ, and they account for unobserved heterogeneity across time and space. Subindex s refers to the fixed effect controlling for the state where the respondent lives, using Mexico City as the base category. Subindex t represents the time fixed effect, using the first quarter of 2016 as the base category. Subindex i represents each individual in the sample. Finally, Φ represents the standard normal cumulative distribution function, while β_0 is the intercept.

3.3 Descriptive statistics

The online Appendix includes all the tables presenting the descriptive statistics of the variables used in the regression analysis. Table 1.2 presents the details of the dependent variables that capture nine types of precarious working conditions in the labour market. Table 1.3 presents descriptive statistics for variables capturing individual and household characteristics. Table 1.4 includes the details of the fixed-effects variables. Table 1.5 presents the details of the 50 economic activities of Mexico's labour market. Table 1.6 specifies the details of the NAICS codes that were used to disaggregate Mexico's labour market into 50 economic activities[13]. Finally, Figure 1.1 presents the descriptive statistics illustrating the informal employment rate and the share of women in each of these 50 economic activities.

4 Results

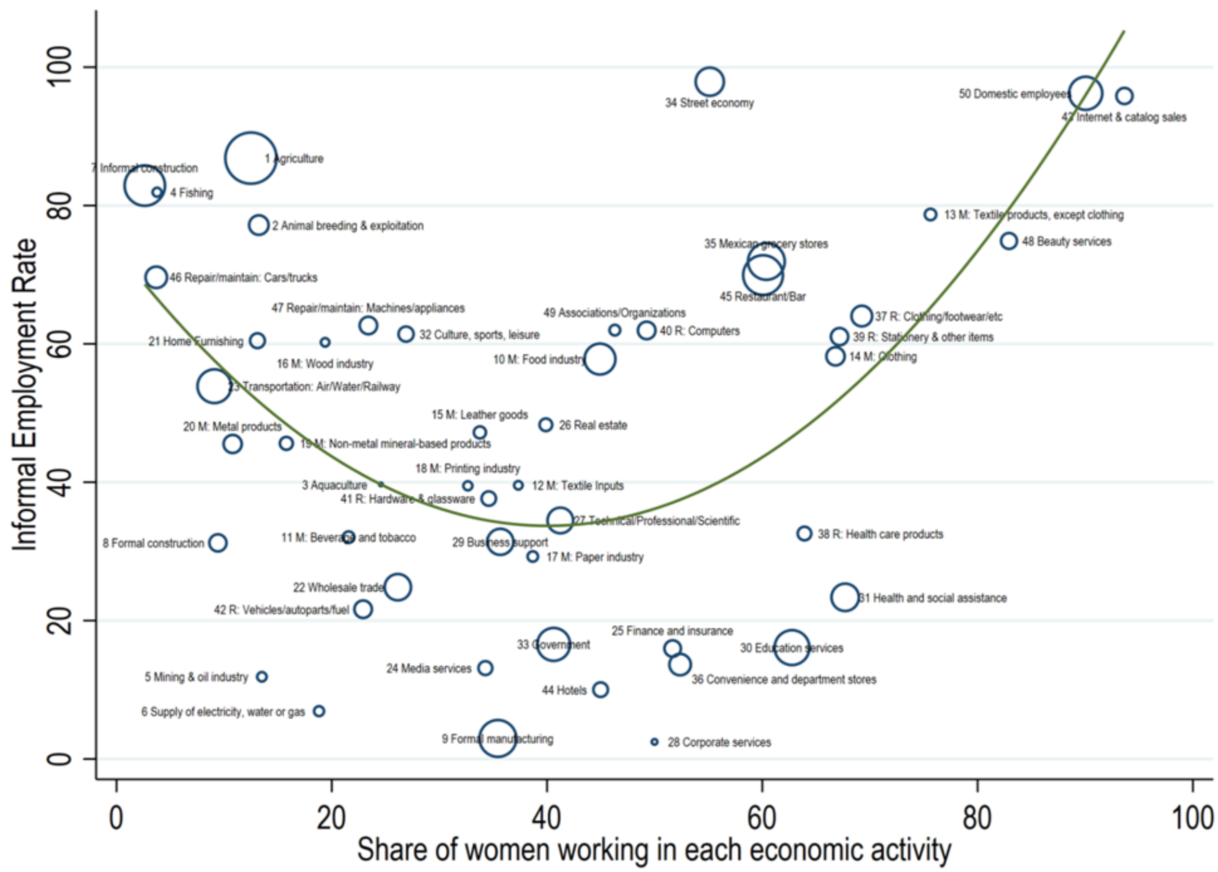
This section begins by presenting the results of the hypothesis test that women are more likely than men to be informal workers. It then examines which precarious working conditions are more likely to be experienced by men and women in the labour market. Finally, it identifies workers who are highly likely to experience specific types of precarious working conditions, depending on the economic activity in which they work. It is worth noting that the results discussed throughout this section capture correlational patterns rather than causal relationships. Therefore, they cannot be considered anything more than merely descriptive evidence.

4.1 Gender, informality and labour precarity

One objective of this research is to evaluate the hypothesis that women are more likely than men to be informal workers. The results obtained from the probit regression evaluating this hypothesis are

[13] *Additional details of the NAICS classification system can also be found, in Spanish, in the INEGI (2015) classification manual.*

Figure 1.1: Informal employment rates and gender composition of the workforce in fifty economic activities (Mexico, 2019)



Note: The size of the dots reflects the share of people working in each economic activity. These 50 economic activities account for 98% of the total labour force in Mexico. Economic activities starting with 'M:' correspond to manufacturing, while those starting with 'R:' correspond to retail. In convenience stores such as 7-Eleven and Oxxo, owners typically hire employees to operate the business, whereas Mexican grocery stores are commonly operated as a form of self-employment. Bachas et al. (2020) also distinguish between 'traditional' stores in Mexico's informal sector and 'modern' stores in the formal sector. The figure shows that gender parity is more likely in economic activities where informality is low (centre of the graph), while gender segmentation is more pronounced in activities with high informality rates (edges of the graph). Men are more likely to dominate informal activities in agriculture and industry, whereas women are more likely to dominate informal activities in services. The figure includes a quadratic fit capturing the relationship between gender composition and informal employment across sectors. The estimates of informal employment rates and workforce composition fall within the coefficients of variation established by INEGI based on the sampling design of the ENOE survey (see Table 1.5 in the online Appendix). A detailed classification of the 50 economic activities is provided in Table 1.6 in the online Appendix.

presented in Column 1 of Table 1.1, and they report a positive and statistically significant coefficient for women. This confirms that, among the working population, Mexican women are more likely to be informal workers relative to men. This is consistent with official statistics indicating that the informal employment rates in Mexico were slightly higher for women (57%) than for men (55%) in 2019 (see online Appendix, Table 1.8).

Nevertheless, this evidence does not imply that women predominate in informal employment. Since the regression analysis is restricted to the working population, the results cannot be generalised to the working-age population. The correct interpretation of the result is that 'working women' in Mexico are more likely to be informally employed than 'working men'. This distinction is crucial after considering national and international statistics on informal employment. In Mexico, the data indicate that in 2019, around 31.5 million people were informally employed, comprising 18.7 million men and 12.7 million women (see online Appendix, Table 1.9). International Labour Organization (2018) also reports that more than two billion people worldwide have informal jobs, comprising around 740 million women and over one billion men.

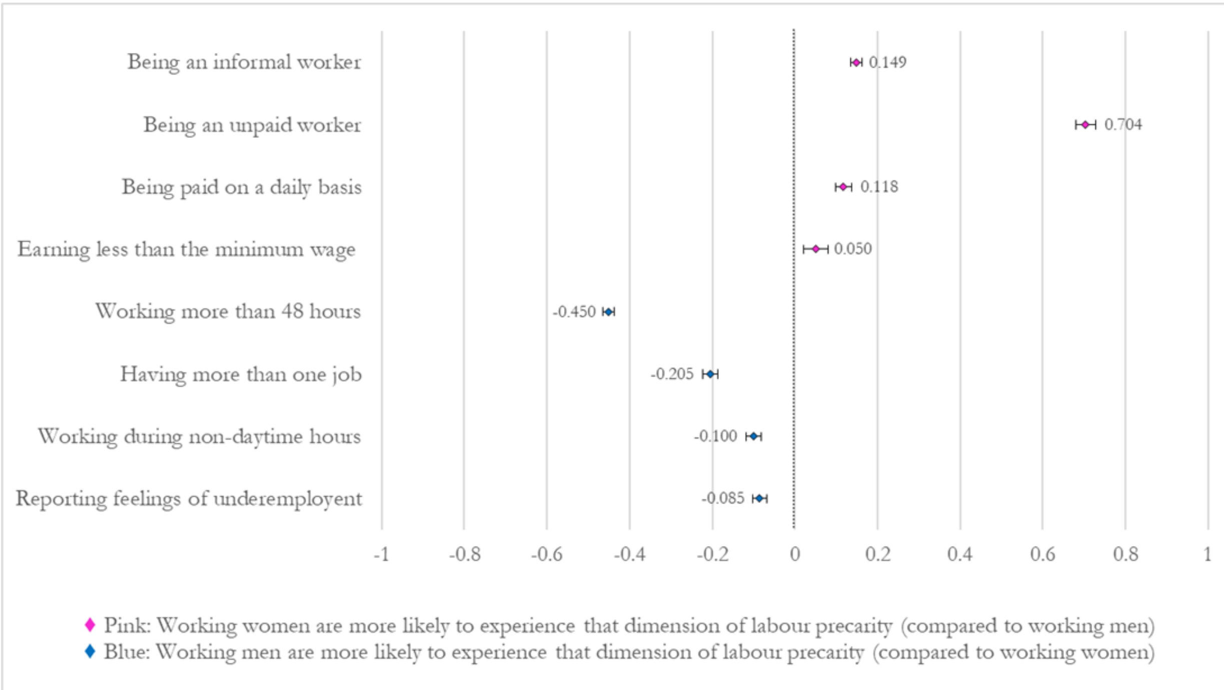
Moreover, Figure 1.6 (see online Appendix) shows that informal employment rates are not higher for women than for men across countries. This research argues that in those countries where informal employment rates are higher for men than for women, a probit regression will typically indicate that working men are more likely to have an informal job than working women. On the other hand, in those countries where informal employment rates are higher for women than for men (such as the case of Mexico), a probit regression will typically report that working women are more likely to have an informal job compared to working men. Nevertheless, the fact that working women in Mexico are more likely to have an informal job does not imply that they predominate in the informal economy.

This interpretation may appear counterintuitive to some readers, but it can be explained with relative simplicity. On one hand, Mexican women show higher informal employment rates than men. On the other hand, there are more Mexican men in the informal workforce in absolute terms. This is not surprising, since men have higher labour force participation rates than women. However, the study highlights both sides of the picture to prevent misinterpretations of the results and to avoid the incorrect conclusion that women predominate in informal jobs.

Beyond this clarification, a major contribution of this research to the analysis of gender and informality is presented in Figure 1.1. This figure shows the informal employment rate and the share of women working in each of these 50 economic activities. The evidence shows that formal economic activities are more likely to exhibit gender balance in the workforce composition. In contrast, highly informal activities are typically gender-segmented, dominated by either men or women. Hence, this analysis explicitly identifies the activities in which women predominate and shows that incorporating informality into the analysis of the gender division of labour offers new insights into this topic.

Another objective of the research has been to identify which precarious working conditions are more likely to be experienced by men and women in the labour market. The results presented in Table 1.1, and illustrated in Figure 1.2, reveal an interesting pattern. Women are more likely to face income-related precarity, such as being unpaid workers, receiving daily wages, or having incomes below the minimum wage. Men, on the other hand, are more likely to face workload-related precarity, such as working more than 48 hours per week, holding two jobs, working during non-daytime hours, and reporting underemployment. These results are consistent with the findings of Hovhannisyan et al. (2022), who studied job quality in developing countries and concluded that men perform better on the income dimension, while women perform better on working conditions.

Figure 1.2: Coefficient plot showing whether men or women are more likely to experience different types of precarious working conditions



Notes: The coefficients are based on the regression results presented in Table 1.1. All coefficients are statistically significant at the 1% level. Positive coefficients indicate that working women are more likely to experience the corresponding dimension of labour precarity compared with working men. Negative coefficients indicate that working men are more likely to experience that dimension of labour precarity. Being a subordinate worker without a contract did not yield statistically significant results and is therefore not shown in this figure.

The results suggest that Mexican men may be more likely to assume the role of household breadwinner. Kanji and Samuel (2017) documented how men endorsing breadwinner roles and conservative views on women's employment are more likely to report feelings of underemployment. Perhaps this could also be associated with their higher likelihood of working more than 48 hours a week or having more than one job. In contrast, women's results suggest that they may juggle multiple responsibilities – such as cooking, cleaning and caregiving – while also contributing to the household financial situation. Hence, Mexican women appear to be assuming the burden of being household jugglers. This may be associated with their tendency towards informal and flexible jobs that are often paid on a daily basis, or their participation as unpaid family workers, helping the family business just to avoid the costs of hiring external labour.

It is essential to emphasise that the results control for individual and household characteristics but, more importantly, for the economic activity in which individuals are employed. Therefore, any observed gender differences in the probability of experiencing a given dimension of precarity are not explained by the over-representation of women in activity X or the predominance of men in activity Y. Instead, controlling for economic activity absorbs such variation, and the results capture gender disparities that persist even within the economic activities. The regressions therefore indicate whether women or men are more likely to experience precarious working conditions even after controlling for the economic activities in which they predominate.

To illustrate this point, take daily wage payments as an example. The results suggest that men working in restaurants are more likely to receive daily pay than women working in financial services – but that reflects sector differences, not gender patterns. A key contribution of the model is showing that, once economic activity is held constant, women are consistently and statistically more likely to be paid on a daily basis than men working in the same activity.

These results contribute to a better understanding of the different challenges that both men and women face in the labour market and the different types of precarious working conditions they must endure. Men are more likely to experience workload precarity, while women are more likely to experience income precarity. These patterns appear consistent with men and women occupying different household roles in Mexico. Men's higher likelihood of experiencing workload precarity may be associated with assuming the role of breadwinner, while women's higher likelihood of experiencing income precarity may be associated with having to juggle various household responsibilities (cooking, cleaning, caregiving) while also contributing to household finances.

4.2 Precarious working conditions across economic activities

The results of this research also contribute to the literature on decent work and labour precarity. The regression outcomes presented in Table 1.1 identify the economic activities where workers are more likely to experience different types of precarious working conditions, while Figure 1.3 illustrates the results by highlighting the top three economic activities in which workers are more likely to experience specific types of precarious working conditions.

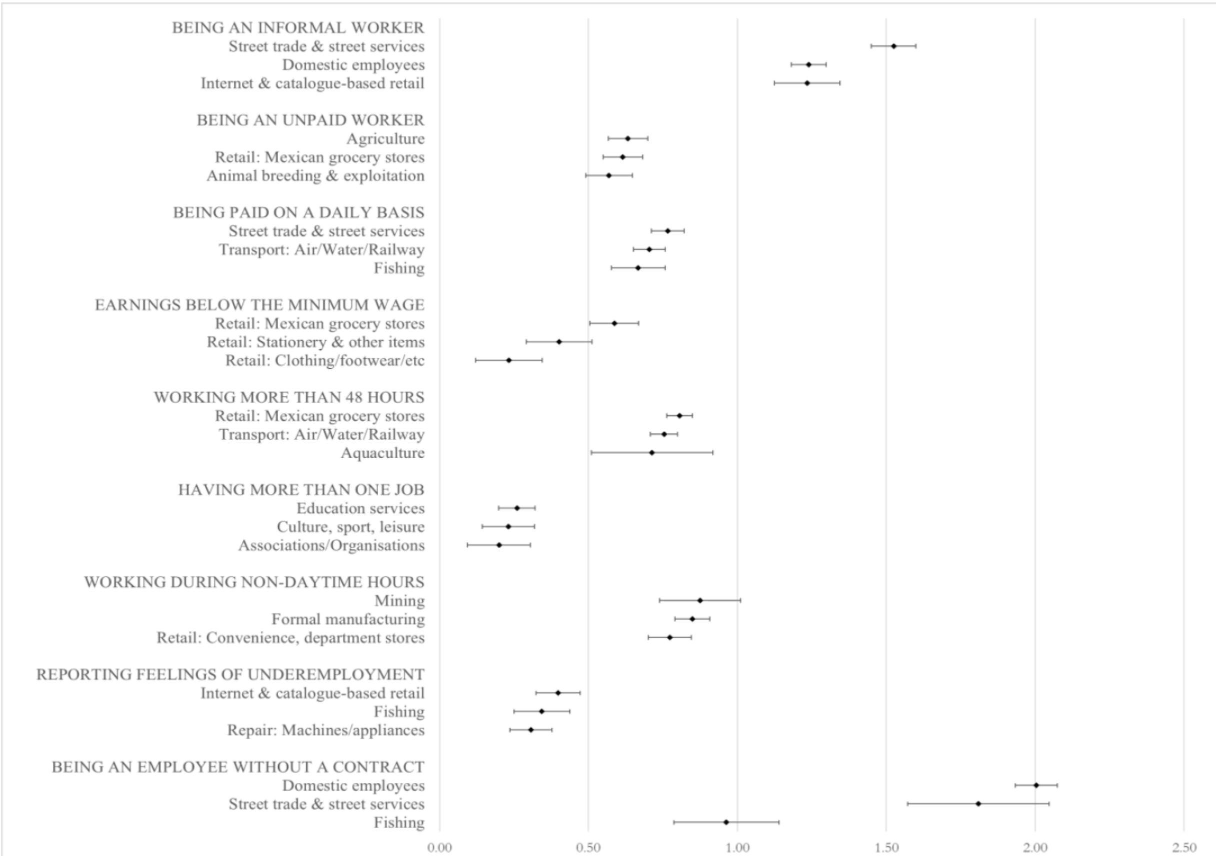
The results show that the probability of being an employee without a contract is particularly high among domestic workers, and in the street economy, in fishing, agriculture, informal construction, beauty salons and auto-repair shops. While it may be the norm for street food businesses not to sign contracts with their employees, not having a contract is more concerning in high-risk activities like residential construction. This raises serious concerns about worker protection and regulatory enforcement for those responsible for building houses in Mexico.

Working more than 48 hours weekly is highly likely among workers in Mexican grocery stores, in transportation, construction and mining, as well as in retail stores selling stationery, clothing or hardware. Unlike convenience stores like 7-Eleven or Oxxo, which hire staff and limit shifts to eight hours, Mexican grocery stores are often family businesses or sources of self-employment, sometimes located in the parking space or first floor of the owner's home, and typically open from early morning until late at night. Regarding transportation services, various newspapers have reported that truck drivers in Mexico endure excessively long shifts, sometimes relying on cocaine and other drugs to stay awake (El Economista, 2010; El Salto, 2022; El Sol de Toluca, 2023). Hence, although regulating working hours among the self-employed in Mexican grocery stores seems unfeasible, the Mexican government could implement regulations to prevent subordinate workers, such as truck drivers, from being compelled to work excessively long shifts.

The probability of holding a second job is higher for workers in education, health, culture, sports, entertainment, organisations, associations and domestic work. Nevertheless, this may not necessarily be considered a situation of labour precarity in all cases. Some jobs in these activities have shorter working hours, which may lead some workers to engage in secondary economic activities to supplement their income.

Receiving daily wages is more likely among street workers, domestic workers and those employed in restaurants and bars, beauty salons, Mexican grocery stores, transportation services, fishing and

Figure 1.3: Coefficient plot showing the top three economic activities in which workers are more likely to experience different types of precarious working conditions



Notes: The coefficients of this figure are based on the regression results presented in Table 1.1. The coefficients derive from probit regressions to estimate the likelihood that workers are experiencing different types of precarious working conditions. All coefficients presented in this figure are statistically significant at the 1% significance level.

agriculture. This type of remuneration may translate into income instability and unpredictable earnings, making people more vulnerable to income shocks and perpetuating labour precarity in Mexico.

Working outside daytime hours is the most widespread dimension of labour precarity, with 28 of the 50 economic activities showing positive and statistically significant results. This pattern indicates that, in Mexico, it is common to work (partially or entirely) outside the 6 am to 8 pm schedule. The regression results show that the highest coefficients are found in mining, formal manufacturing, hotels, government jobs, and department and convenience stores. In manufacturing, production processes often operate continuously, requiring staff to work night shifts between 8 pm and 6 am. A similar situation is found in hotels and some convenience or department stores. In hotels, personnel are needed at the front desk throughout the night, while in convenience stores, employees may work until closing time (often 10 or 11 pm) or cover overnight shifts if the stores operate 24 hours a day.

Reporting feelings of underemployment refers to individuals who state that they would like to work more hours than they currently do. This is more likely to be observed in catalogue or online sales (an activity where women predominate), as well as in repairing machinery and appliances, among workers in the street economy and among those engaged in fishing. Finally, earning less than the minimum wage despite working 48 hours is more likely among individuals employed in Mexican grocery stores, stationery stores, clothing stores, and in animal breeding and exploitation.

These findings reveal that precarious working conditions are widespread across Mexico's labour market. Table 1.7 (see online Appendix) presents additional insights into precarious working conditions throughout Mexican labour markets. In sum, the evidence of this research identifies precarious working conditions in specific economic activities, making it easier to understand how vulnerable workers are distributed throughout the labour market and which economic activities require the most urgent attention from policy makers and labour rights advocates. Effective policy interventions must be designed with consideration for the characteristics of each economic activity. This nuanced understanding of how labour precarity operates across Mexico's labour markets provides a foundation for targeted reforms that can meaningfully improve working conditions for the country's most vulnerable workers.

5 Discussions

The findings of this research contribute to a better understanding of the informal economy, to the analysis of other types of precarious working conditions, and to how women and men face these labour market challenges. Nevertheless, one aspect of this study may be seen as controversial rather than insightful. This research states that women do not predominate in informal employment and that they do not have higher informal employment rates across countries. Therefore, it is relevant to include a brief discussion about this claim.

Researchers and international organisations may have framed the informal economy as a space dominated by women without realising that their statistics were incomplete. For a long time, countries measured informal employment rates by only considering industry and services. International Labour Organization (2013) explained that agricultural activities were typically not considered because including them would demand "a considerable expansion of survey operations and increase in costs". International Labour Organization (2002) also explained that at the beginning of the 21st century, Mexico, India, and South Africa were among the few countries considering agriculture in their estimates of informal employment rates.

According to The Sustainable Development Goals Report (2021), the informal employment rate for agricultural activities worldwide was 90.7%, while for non-agricultural activities it was 48.9%. Since most agricultural workers are informally employed, focusing on capturing heterogeneity and fluctuations in informal employment in industry and services is justifiable. Nevertheless, most researchers overlook the fact that men constitute the majority of the agricultural workforce in their countries. Figure 1.5 (see online Appendix) shows that the agricultural workforce is predominantly male in all regions of the world. Similarly, Doss and Gottlieb (2025) also find that men make up the majority of the agricultural workforce in most countries.

International Labour Organization (2018) recognised the need to estimate labour informality without excluding agricultural activities, noting that the SDG indicator 8.3.1 was only measuring the informal employment rate in non-agricultural activities. Acting on this recommendation, the indicator was revised to include informal employment in all sectors, including agriculture (Sustainable Development Goals, 2020). The previous SDG indicator was "8.3.1: Proportion of informal employment in non-agricultural employment, by sex". The updated indicator is "8.3.1: Proportion

Table 1.1: Correlates of precarious working conditions

	(1) Being an informal worker		(2) Subordinate without contract		(3) Being an unpaid worker	
	Coeff.	*Std. Err.*	*Coeff.*	*Std. Err.*	*Coeff.*	*Std. Err.*
Women (Compared to men)	**0.149*** **	(0.007)	-0.001	(0.009)	**0.704*** **	(0.012)
1 Agriculture	**0.212*** **	(0.020)	**0.753*** **	(0.026)	**0.633*** **	(0.034)
2 Animal breeding & exploitation	**0.046* **	(0.027)	**0.264*** **	(0.035)	**0.569*** **	(0.040)
3 Aquaculture	-0.766***	(0.107)	-0.099	(0.119)	-0.051	(0.230)
4 Fishing	**0.355*** **	(0.051)	**0.963*** **	(0.090)	**0.166** **	(0.073)
5 Mining	-1.475***	(0.044)	-1.277***	(0.050)	-0.893***	(0.132)
6 Supply of electricity, water or gas	-1.792***	(0.062)	-1.552***	(0.064)	-0.971***	(0.181)
7 Informal construction	**0.503*** **	(0.020)	**0.669*** **	(0.024)	-0.890***	(0.054)
8 Formal construction	-0.829***	(0.028)	-0.527***	(0.030)	-0.795***	(0.108)
9 Formal manufacturing	-2.136***	(0.027)	-1.698***	(0.027)	-1.209***	(0.068)
10 Food industry	-0.447***	(0.021)	-0.348***	(0.025)	**0.170*** **	(0.037)
11 Beverage and tobacco industry	-0.939***	(0.039)	-0.681***	(0.043)	-0.176*	(0.107)
12 Textile Inputs	-1.193***	(0.049)	-1.556***	(0.077)	-0.532***	(0.087)
13 Textile products, except clothing	-0.081**	(0.041)	-0.591***	(0.066)	-0.369***	(0.067)
14 Clothing manufacturing	-0.497***	(0.027)	-0.473***	(0.033)	-0.498***	(0.061)
15 Leather goods manufacturing	-0.574***	(0.031)	-0.178***	(0.036)	-0.451***	(0.073)
16 Wood industry	-0.216***	(0.047)	0.020	(0.058)	**0.214*** **	(0.072)
17 Paper industry	-1.224***	(0.049)	-1.235***	(0.069)	-0.277***	(0.093)
18 Printing and related industries	-0.588***	(0.050)	-0.250***	(0.060)	-0.244**	(0.099)
19 Non-metal mineral-based products	-0.612***	(0.031)	-0.499***	(0.036)	**0.148*** **	(0.051)
20 Metal products	-0.515***	(0.026)	-0.301***	(0.031)	-0.283***	(0.057)
21 Furniture, mattresses, and blinds	-0.174***	(0.031)	-0.053	(0.037)	-0.129**	(0.063)
22 Wholesale trade	-0.931***	(0.023)	-0.660***	(0.026)	-0.311***	(0.048)
23 Transport: Air/Water/Railway	-0.169***	(0.021)	-0.029	(0.024)	-0.846***	(0.066)
24 Media services	-1.344***	(0.042)	-1.060***	(0.054)	-1.160***	(0.118)
25 Finance and insurance	-1.270***	(0.036)	-1.200***	(0.039)	-1.501***	(0.184)
26 Real estate	-0.300***	(0.036)	-0.033	(0.044)	-0.198**	(0.079)
27 Technical/Professional/Scientific	-0.550***	(0.024)	-0.250***	(0.030)	-0.689***	(0.054)
28 Corporate services	-1.791***	(0.117)	-1.361***	(0.153)	-1.274***	(0.251)
29 Business support	-0.893***	(0.023)	-0.994***	(0.027)	-0.609***	(0.054)
30 Education services	-1.197***	(0.022)	-1.347***	(0.027)	-0.913***	(0.050)
31 Health and social assistance	-0.975***	(0.023)	-0.921***	(0.027)	-0.540***	(0.043)
32 Culture, sports, leisure	0.004	(0.032)	0.051	(0.037)	-0.543***	(0.071)
33 Government	-1.262***	(0.021)	-1.514***	(0.027)	-0.754***	(0.049)
34 Street trade & street services	**1.525*** **	(0.038)	**1.809*** **	(0.121)	**0.409*** **	(0.037)
35 R: Mexican grocery stores	**0.038* **	(0.021)	**0.630*** **	(0.029)	**0.616*** **	(0.034)
36 R: Convenience, department	-1.613***	(0.027)	-1.322***	(0.030)	-0.807***	(0.057)
37 R: Clothing/footwear/etc	-0.115***	(0.027)	-0.011	(0.034)	-0.008	(0.046)
38 R: Health care products	-0.823***	(0.038)	-0.620***	(0.045)	-0.248***	(0.077)
39 R: Stationery & other items	-0.102***	(0.030)	**0.140*** **	(0.041)	**0.283*** **	(0.045)
40 R: Computers, household items	-0.076***	(0.028)	-0.170***	(0.038)	**0.136*** **	(0.048)
41 R: Hardware & glassware	-0.563***	(0.033)	-0.279***	(0.039)	**0.149** **	(0.061)
42 R: Vehicles/autoparts/fuel	-1.000***	(0.030)	-0.770***	(0.033)	-0.465***	(0.062)
43 Internet & catalog-based retail	**1.235*** **	(0.056)	-0.380***	(0.120)	-0.905***	(0.073)
44 Hotels	-1.563***	(0.035)	-1.210***	(0.039)	-1.145***	(0.091)
45 Restaurant/Bar: Food/drinks	**0.047** **	(0.020)	**0.237*** **	(0.023)	**0.211*** **	(0.034)
46 Repair: Cars/trucks	**0.173*** **	(0.025)	**0.684*** **	(0.033)	-0.173***	(0.051)
47 Repair: Machines/appliances	0.033	(0.028)	-0.122***	(0.041)	-0.153***	(0.054)
48 Haircuts/Grooming/Makeup	**0.239*** **	(0.034)	**0.847*** **	(0.055)	-1.250***	(0.097)
49 Associations/Organizations	**0.071* **	(0.041)	**0.180*** **	(0.042)	-0.308***	(0.083)
50 Domestic employees	**1.240*** **	(0.030)	**2.004*** **	(0.036)	-	-
51 Other economic activities	-	-	-	-	-	-
Control variables	Yes		Yes		Yes	
Age	Yes		Yes		Yes	
Age squared	Yes		Yes		Yes	
Marital status	Yes		Yes		Yes	
Level of education	Yes		Yes		Yes	
Household location: Rural /Urban	Yes		Yes		Yes	
Household socioeconomic strata	Yes		Yes		Yes	
Household members	Yes		Yes		Yes	
Number of kids in the household	Yes		Yes		Yes	
Year fixed effects	Yes		Yes		Yes	
State fixed effects	Yes		Yes		Yes	
Constant	2.742***	(0.041)	2.689***	(0.054)	-0.768***	(0.084)
Observations	687,955		479,721		657,630	

Table 1.1: Correlates of precarious working conditions (continued)

	(4) Being paid on a daily basis		(5) Earning less than the minimum wage		(6) Working more than 48 hours	
	Coeff.	*Std. Err.*	*Coeff.*	*Std. Err.*	*Coeff.*	*Std. Err.*
Women (Compared to men)	**0.118*****	(0.010)	**0.050*****	(0.015)	**-0.450*****	(0.007)
1 Agriculture	**0.396*****	(0.025)	-0.254***	(0.043)	-0.422***	(0.022)
2 Animal breeding & exploitation	-0.331***	(0.034)	**0.216*****	(0.052)	**0.229*****	(0.027)
3 Aquaculture	-0.608***	(0.231)	-0.142	(0.258)	**0.714*****	(0.104)
4 Fishing	**0.667*****	(0.046)	-0.140	(0.092)	-0.203***	(0.046)
5 Mining	-1.099***	(0.086)	-0.552***	(0.136)	**0.330*****	(0.039)
6 Supply of electricity, water or gas	-1.589***	(0.147)	-0.502***	(0.130)	-0.364***	(0.053)
7 Informal construction	-0.434***	(0.026)	-0.538***	(0.052)	**0.172*****	(0.021)
8 Formal construction	-0.882***	(0.050)	-0.557***	(0.085)	**0.400*****	(0.029)
9 Formal manufacturing	-1.345***	(0.037)	-0.548***	(0.053)	-0.079***	(0.023)
10 Food industry	**0.082*****	(0.027)	-0.103**	(0.048)	**0.218*****	(0.024)
11 Beverage and tobacco industry	-0.404***	(0.054)	0.025	(0.104)	**0.290*****	(0.040)
12 Textile Inputs	-1.186***	(0.089)	-0.308***	(0.100)	-0.446***	(0.072)
13 Textile products, except clothing	-1.477***	(0.084)	-0.201**	(0.079)	-0.392***	(0.047)
14 Clothing manufacturing	-1.185***	(0.045)	-0.069	(0.061)	**0.198*****	(0.029)
15 Leather goods manufacturing	-1.280***	(0.082)	-0.357***	(0.105)	**0.384*****	(0.032)
16 Wood industry	-0.523***	(0.072)	-0.207**	(0.095)	**0.117****	(0.047)
17 Paper industry	-1.109***	(0.108)	-0.357***	(0.107)	-0.136***	(0.052)
18 Printing and related industries	-0.898***	(0.150)	-0.502***	(0.161)	**0.102***	(0.056)
19 Non-metal mineral-based products	-0.688***	(0.051)	-0.175**	(0.078)	0.040	(0.035)
20 Metal products	-0.873***	(0.044)	-0.339***	(0.080)	**0.081*****	(0.028)
21 Furniture, mattresses, and blinds	-0.876***	(0.055)	-0.248***	(0.073)	**0.092*****	(0.032)
22 Wholesale trade	-0.725***	(0.037)	-0.231***	(0.060)	**0.385*****	(0.024)
23 Transport: Air/Water/Railway	**0.705*****	(0.027)	-0.010	(0.048)	**0.754*****	(0.023)
24 Media services	-1.248***	(0.090)	-0.359***	(0.118)	0.050	(0.038)
25 Finance and insurance	-1.687***	(0.124)	-0.400***	(0.117)	**0.237*****	(0.034)
26 Real estate	-0.317***	(0.055)	-0.359***	(0.119)	-0.005	(0.039)
27 Technical/Professional/Scientific	-0.757***	(0.049)	-0.637***	(0.095)	-0.193***	(0.028)
28 Corporate services	-2.144***	(0.305)	-0.880***	(0.287)	0.143*	(0.085)
29 Business support	-0.350***	(0.030)	-0.084	(0.056)	**0.271*****	(0.024)
30 Education services	-1.286***	(0.047)	-0.963***	(0.104)	-0.966***	(0.029)
31 Health and social assistance	-0.829***	(0.044)	-0.495***	(0.069)	-0.375***	(0.028)
32 Culture, sports, leisure	0.012	(0.040)	-0.323***	(0.087)	-0.273***	(0.038)
33 Government	-1.726***	(0.064)	-0.669***	(0.073)	-0.059**	(0.023)
34 Street trade & street services	**0.767*****	(0.028)	0.082*	(0.050)	-0.044*	(0.025)
35 R: Mexican grocery stores	**0.376*****	(0.026)	**0.587*****	(0.042)	**0.806*****	(0.022)
36 R: Convenience, department	-0.387***	(0.035)	-0.235***	(0.059)	**0.249*****	(0.030)
37 R: Clothing/footwear/etc	-0.147***	(0.038)	**0.233*****	(0.057)	**0.480*****	(0.028)
38 R: Health care products	-0.710***	(0.064)	-0.078	(0.091)	**0.257*****	(0.040)
39 R: Stationery & other items	-0.087**	(0.043)	**0.402*****	(0.056)	**0.448*****	(0.031)
40 R: Computers, household items	**0.158*****	(0.038)	0.033	(0.059)	**0.197*****	(0.030)
41 R: Hardware & glassware	-0.581***	(0.061)	-0.054	(0.083)	**0.397*****	(0.035)
42 R: Vehicles/autoparts/fuel	-0.739***	(0.056)	-0.057	(0.066)	**0.340*****	(0.029)
43 Internet & catalog-based retail	-1.383***	(0.078)	-0.677***	(0.115)	-0.981***	(0.055)
44 Hotels	-0.929***	(0.056)	-0.434***	(0.075)	**0.108*****	(0.032)
45 Restaurant/Bar: Food/drinks	**0.391*****	(0.025)	-0.061	(0.046)	**0.252*****	(0.022)
46 Repair: Cars/trucks	-0.289***	(0.033)	0.096	(0.060)	**0.322*****	(0.026)
47 Repair: Machines/appliances	-0.205***	(0.043)	-0.210***	(0.066)	-0.079***	(0.031)
48 Haircuts/Grooming/Makeup	**0.178*****	(0.041)	0.051	(0.071)	**0.072****	(0.034)
49 Associations/Organizations	-0.497***	(0.067)	0.016	(0.100)	-0.006	(0.044)
50 Domestic employees	**0.548*****	(0.026)	-0.258***	(0.049)	-0.303***	(0.026)
51 Other economic activities	-	-	-	-	-	-
Control variables	Yes		Yes		Yes	
Age	Yes		Yes		Yes	
Age squared	Yes		Yes		Yes	
Marital status	Yes		Yes		Yes	
Level of education	Yes		Yes		Yes	
Household location: Rural /Urban	Yes		Yes		Yes	
Household socioeconomic strata	Yes		Yes		Yes	
Household members	Yes		Yes		Yes	
Number of kids in the household	Yes		Yes		Yes	
Year fixed effects	Yes		Yes		Yes	
State fixed effects	Yes		Yes		Yes	
Constant	0.045	(0.047)	-2.062***	(0.078)	-1.165***	(0.038)
Observations	486,814		687,955		682,555	

Table 1.1: Correlates of precarious working conditions (continued)

	(7) Having more than one job		(8) Working during non-daytime hours		(9) Reporting underemployent	
	Coeff.	*Std. Err.*	*Coeff.*	*Std. Err.*	*Coeff.*	*Std. Err.*
Women (Compared to men)	**-0.205*****	(0.009)	**-0.100*****	(0.009)	**-0.085*****	(0.009)
1 Agriculture	0.047	(0.028)	-0.086***	(0.027)	-0.043*	(0.025)
2 Animal breeding & exploitation	-0.079**	(0.036)	**0.287*****	(0.039)	-0.344***	(0.036)
3 Aquaculture	-0.617***	(0.181)	0.201	(0.169)	-0.375**	(0.162)
4 Fishing	-0.115**	(0.056)	-0.079	(0.071)	**0.344*****	(0.048)
5 Mining	-0.283***	(0.057)	**0.875*****	(0.069)	-0.594***	(0.062)
6 Supply of electricity, water or gas	-0.257***	(0.054)	**0.441*****	(0.044)	-0.873***	(0.077)
7 Informal construction	-0.140***	(0.030)	-0.223***	(0.027)	**0.107*****	(0.026)
8 Formal construction	-0.232***	(0.050)	**0.132*****	(0.039)	-0.264***	(0.039)
9 Formal manufacturing	-0.456***	(0.032)	**0.849*****	(0.030)	-0.643***	(0.030)
10 Food industry	-0.146***	(0.033)	**0.221*****	(0.031)	-0.231***	(0.030)
11 Beverage and tobacco industry	-0.372***	(0.057)	**0.473*****	(0.055)	-0.510***	(0.057)
12 Textile Inputs	0.028	(0.065)	**0.452*****	(0.090)	-0.340***	(0.074)
13 Textile products, except clothing	-0.208***	(0.055)	0.036	(0.050)	-0.092*	(0.052)
14 Clothing manufacturing	-0.163***	(0.042)	0.052	(0.038)	-0.129***	(0.038)
15 Leather goods manufacturing	-0.409***	(0.067)	**0.080***	(0.046)	-0.338***	(0.050)
16 Wood industry	-0.125*	(0.067)	0.002	(0.064)	-0.097	(0.065)
17 Paper industry	-0.288***	(0.075)	**0.536*****	(0.072)	-0.382***	(0.072)
18 Printing and related industries	-0.260***	(0.073)	0.043	(0.072)	-0.113	(0.074)
19 Non-metal mineral-based products	-0.260***	(0.049)	**0.230*****	(0.053)	-0.211***	(0.050)
20 Metal products	-0.242***	(0.042)	-0.122***	(0.040)	0.032	(0.037)
21 Furniture, mattresses, and blinds	-0.122***	(0.047)	-0.149***	(0.043)	0.003	(0.043)
22 Wholesale trade	-0.313***	(0.036)	**0.178*****	(0.032)	-0.364***	(0.034)
23 Transport: Air/Water/Railway	-0.331***	(0.033)	**0.376*****	(0.033)	-0.179***	(0.029)
24 Media services	-0.114**	(0.055)	**0.542*****	(0.046)	-0.403***	(0.059)
25 Finance and insurance	-0.268***	(0.053)	**0.369*****	(0.035)	-0.516***	(0.052)
26 Real estate	-0.124**	(0.056)	0.042	(0.053)	-0.008	(0.048)
27 Technical/Professional/Scientific	-0.106***	(0.041)	0.052	(0.035)	-0.103***	(0.036)
28 Corporate services	-0.424***	(0.107)	**0.354*****	(0.075)	-0.662***	(0.148)
29 Business support	-0.133***	(0.035)	**0.441*****	(0.036)	-0.185***	(0.032)
30 Education services	**0.260*****	(0.031)	**0.308*****	(0.028)	-0.556***	(0.031)
31 Health and social assistance	**0.073****	(0.033)	**0.396*****	(0.035)	-0.356***	(0.034)
32 Culture, sports, leisure	**0.231*****	(0.045)	**0.237*****	(0.057)	**0.079***	(0.042)
33 Government	-0.029	(0.031)	**0.666*****	(0.029)	-0.802***	(0.034)
34 Street trade & street services	-0.082**	(0.035)	-0.207***	(0.031)	**0.211*****	(0.028)
35 R: Mexican grocery stores	-0.206***	(0.032)	0.043	(0.029)	-0.058**	(0.028)
36 R: Convenience, department	-0.336***	(0.043)	**0.774*****	(0.037)	-0.604***	(0.040)
37 R: Clothing/footwear/etc	-0.262***	(0.047)	-0.105***	(0.036)	-0.027	(0.037)
38 R: Health care products	-0.301***	(0.063)	**0.285*****	(0.060)	-0.127**	(0.058)
39 R: Stationery & other items	-0.183***	(0.052)	-0.128***	(0.041)	-0.005	(0.041)
40 R: Computers, household items	-0.272***	(0.048)	-0.096***	(0.037)	-0.008	(0.038)
41 R: Hardware & glassware	-0.399***	(0.057)	0.034	(0.047)	-0.191***	(0.054)
42 R: Vehicles/autoparts/fuel	-0.326***	(0.051)	**0.316*****	(0.043)	-0.342***	(0.045)
43 Internet & catalog-based retail	-0.019	(0.051)	-0.504***	(0.040)	**0.398*****	(0.038)
44 Hotels	-0.419***	(0.048)	**0.720*****	(0.059)	-0.629***	(0.057)
45 Restaurant/Bar: Food/drinks	-0.099***	(0.031)	**0.282*****	(0.030)	-0.119***	(0.027)
46 Repair: Cars/trucks	-0.313***	(0.038)	-0.275***	(0.033)	**0.127*****	(0.031)
47 Repair: Machines/appliances	-0.130***	(0.048)	-0.418***	(0.038)	**0.307*****	(0.036)
48 Haircuts/Grooming/Makeup	-0.016	(0.044)	-0.461***	(0.042)	**0.276*****	(0.042)
49 Associations/Organizations	**0.200*****	(0.054)	**0.520*****	(0.056)	-0.441***	(0.067)
50 Domestic employees	**0.092*****	(0.031)	**0.142*****	(0.028)	-0.216***	(0.029)
51 Other economic activities	-	-	-	-	-	-
Control variables	Yes		Yes		Yes	
Age	Yes		Yes		Yes	
Age squared	Yes		Yes		Yes	
Marital status	Yes		Yes		Yes	
Level of education	Yes		Yes		Yes	
Household location: Rural /Urban	Yes		Yes		Yes	
Household socioeconomic strata	Yes		Yes		Yes	
Household members	Yes		Yes		Yes	
Number of kids in the household	Yes		Yes		Yes	
Year fixed effects	Yes		Yes		Yes	
State fixed effects	Yes		Yes		Yes	
Constant	-2.193***	(0.053)	-1.718***	(0.049)	-1.215***	(0.048)
Observations	686,104		671,537		687,955	

of informal employment in total employment, by sector and sex".

International Labour Organization (2018) released a report that harmonised micro-datasets of labour force surveys from 112 countries, covering nearly 90% of the global workforce. For the first time, this report includes the estimation of informal employment rates across countries, considering not only industry and services, but also agricultural activities. My paper presents an analysis of the statistics presented by International Labour Organization (2018) in Appendix B.1 of their report. Figure 1.6 (see online Appendix) presents an interesting cross-country analysis of informal employment rates (including agriculture). The initial takeaway from this figure is that half the countries report higher informal employment rates for men. Although this is a relevant finding, several aspects of this figure warrant detailed discussion. To preserve the analytical coherence of this section, these points are addressed in the notes beneath the figure. However, the figure shows that, once the agricultural sector is considered, there is no evidence to suggest that women have systematically higher informal employment rates than men across countries.

According to World Bank Open Data (2025), 26% of the global labour force was working in agriculture during 2023. This research also highlights that most agricultural workers are men. Hence, excluding agricultural activities results in omitting some women and several men who work informally. This omission skews the estimates of informal employment rates, leading to overestimations and misinterpretations about gender disparities in informal labour markets. Doing so is the first step to building more reliable statistics of the informal economy and gaining a deeper understanding of how this heterogeneous phenomenon affects men and women differently.

6 Conclusion

This research advances the study of the informal economy by demonstrating that it is not a monolithic block of vulnerable workers but rather a deeply heterogeneous phenomenon whose complexity extends well beyond the formal-informal dichotomy. By leveraging official labour force surveys, this study estimates informality rates across fifty distinct economic activities with an unprecedented level of detail, including the size of Mexico's street economy. As a result, this research challenges the widespread belief that informal economic activities are invisible or elusive from official statistics. The research further contributes to the study of labour precarity by adopting a multidimensional lens, rather than collapsing its many dimensions into a composite index. The central lesson for policymakers and labour rights advocates is clear: one-size-fits-all interventions are destined to fall short. Effective reforms demand tailored policies identifying the precarity and vulnerability of each activity, such as promoting contract enforcement in residential construction or promoting shift-length regulation for truck drivers. Moreover, researchers working in other national contexts can draw inspiration from this research to study the heterogeneity of their own informal economies and the prevalence of precarious working conditions in their labour markets.

The research also yields three previously undocumented findings on the gender division of labour. First, informal economic activities exhibit striking gender segmentation: women constitute the overwhelming majority in activities such as domestic work and beauty services, while men predominate in others such as residential construction and agriculture. Formal activities, by contrast, tend towards a considerably more balanced workforce composition. Potential explanations for this pattern include the interplay between risk exposure and lack of social protection in informal jobs, which may reinforce traditional gender norms that assign physically demanding or hazardous tasks to men. Meanwhile, female-dominated informal activities reveal how the longstanding confinement of women to looking good and cleaning the household has been transposed into the labour market—with women now offering precisely these skills as paid work in other households or commercial settings. Second, this research documents that working men are more likely to experience workload-related precarity, arguably driven by internalised breadwinner norms that lead them to accept excessive hours, while working women, conversely, are more likely to face income-related precarity, reflecting a dual burden whereby primary responsibility for the household is combined with the need to also make financial contributions to the household. Third, this study challenges the prevailing narrative that women predominate in the informal economy. Whilst working women in Mexico report slightly higher informality rates than their male counterparts, men constitute the majority of the informal workforce in absolute terms. Cross-country evidence further demonstrates that women do not systematically exhibit higher informal employment rates than men once the agricultural sector is included. This finding also introduces a methodological nuance: probit regressions capture the relative likelihood of working informally among those who participate in the labour market, and therefore mirror each country's underlying informality rates by gender rather than constituting evidence of female predominance in informal jobs.

To conclude, I raise some constructive criticisms to the Sustainable Development Goals. The SDGs provide clear, measurable targets for critical development issues like poverty and inequality, yet they lack a specific goal for reducing informal employment rates. SDG 1 explicitly aims to halve the proportion of people living in poverty, while SDG 10 aims to achieve sustained income growth for the bottom 40% of the population at a rate higher than the national average. Nevertheless, SDG 8 lacks a target to reduce informal employment rates in developing countries. Instead, it vaguely encourages the formalisation and growth of micro-, small- and medium-sized enterprises. However, this formulation is problematic as it addresses enterprise formalisation rather than workers' access to formal employment. As previously explained, several informal workers are employed by formally registered enterprises. Given that informal employment remains the predominant form of work in developing economies and is intrinsically linked to poverty and inequality, this represents a significant oversight. A concrete target–such as halving informality rates in developing countries by 2030–would have provided a measurable direction for policy interventions and international development efforts. This omission constitutes a missed opportunity to address one of the most pressing challenges facing labour markets in the Global South. Hopefully, this criticism may contribute to making the reduction of informal employment rates an international priority in the future.

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